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Key Points:

- We introduce an idealized but computationally efficient method to estimate transit-time distributions (TTDs) from climate-model archives
- In ACCESS-ESM1.5 it takes up to 1300 ± 350 years for 10% of the water on the North Pacific seabed to reach the ocean surface
- Under SSP3-7.0 climate change a slower future ocean extends sequestration times by $\sim 30\%$ which exceeds $\sim 20\%$ internal climate variability

Supporting Information:

Supporting Information may be found in the online version of this article.

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The Sequestration Efficiency of the Deep Ocean

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Abstract The abyssal ocean may provide adequate sequestration for Carbon Capture and Storage and Carbon Dioxide Removal. The transit of carbon from seafloor release to ocean surface can take millennia, as it occurs through many water pathways characterized by long-tailed transit-time distributions (TTDs). However, multi-millennial simulations of TTDs are typically prohibitively expensive. Here, we introduce an idealized but computationally efficient methodological framework for computing TTDs offline from climate-model archives. We applied this framework to one Earth System Model to estimate the deep ocean sequestration efficiency for the 2030s and 2090s ocean circulations. We found that sequestration is most efficient on abyssal plains isolated from deep branches of the conveyor belt, such as the North Pacific Basins, where less than 10% of the water reaches the surface after 1,000 years. The climate-warming-induced slowdown of the 2090s deep-ocean circulation extends return times by about 30%, which exceeds internal climate variability ($\sim 20\%$).

Plain Language Summary To limit global warming, we must reduce emissions and actively remove CO₂ from the atmosphere. For both Carbon Dioxide Removal and Carbon Storage and Capture at the source, the abyssal ocean may provide adequate long-term storage. This is because dissolved carbon can take up to millennia to travel from the seabed to the surface. We could compute these seabed-to-surface transit times using multi-millennial climate-model simulations, but this is expensive and time-consuming, possibly taking years on powerful supercomputers. To compute these transit times, we introduce a highly efficient method that uses only climate-model archives; that is, without having to run an expensive climate model. While our method is not a perfect substitute for full-climate-model simulations, it allowed us to gain new insights from a wide range of simulations for a substantially smaller fraction of the time and resources. We applied our method to ACCESS-ESM1.5, the Australian climate model, and identified preferred locations for highly efficient sequestration such as the North Pacific Basins, where less than 10% of carbon will return to the surface after 1,000 years. We also found that global warming weakens the deep ocean 2090s circulation and tends to extend sequestration by about 30%.

1. Introduction

As greenhouse gas emissions into the atmosphere continue to rise, limiting global warming to less than 2°C cannot be achieved by only reducing carbon dioxide (CO₂) emissions. Carbon Dioxide Removal (CDR) and Carbon Capture and Storage (CCS) technologies are required (e.g., Gattuso et al., 2021; Riahi et al., 2023). The ocean currently holds about 50 times more carbon than the atmosphere (DeVries, 2022) and could thus be used for CDR and CCS storage (Doney et al., 2024). But while the ocean has absorbed about 25% of anthropogenic CO₂ and will naturally absorb more in the future, this process is slow relative to the pace of climate change (Wong & Matear, 1998).

Many CDR and CCS strategies that leverage the ocean's potential for storing carbon have been proposed, including fertilizing the ocean (Pessarrodona et al., 2023), growing seaweed (DeAngelo et al., 2023; Hurd et al., 2022; Ross et al., 2023), artificial upwelling or downwelling (Oschlies et al., 2010), increasing alkalinity (Lenton et al., 2018), and injecting carbon into the deep ocean (Marchetti, 1977; Oschlies et al., 2025) or the sediments (Herzog, 2001; Ringrose, 2020). Key to effective storage is the carbon sequestration time, that is, the

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time that the removed/captured carbon is sequestered before being released into the atmosphere. In the ocean, this sequestration time is controlled by the time that water itself takes to return to the ocean surface.

The time water takes to reach the surface has been estimated in a number of contexts, using various models and configurations (e.g., DeVries & Holzer, 2019; DeVries & Primeau, 2011; DeVries et al., 2012; Haine et al., 2008, 2025; Holzer & Hall, 2000; Holzer et al., 2020, 2021; Nowicki et al., 2024; Pasquier, Holzer, & Chamberlain, 2024; Primeau, 2005; Primeau & Holzer, 2006; Siegel et al., 2021). Yet, the effects from climate change and internal climate variability have not been investigated. This is likely because it is complicated to directly simulate in most ocean models. While logging the return times of a dye injected in the ocean interior is feasible, doing so for many injection dates and locations and for multi-millennial simulation times (Primeau, 2005) quickly becomes computationally prohibitive. A state-of-the-art and computationally efficient workaround is to fix the ocean circulation and track water backwards in time, that is, from the surface back into the ocean interior, using what is called an adjoint boundary propagator (e.g., Haine et al., 2025; Holzer & Hall, 2000; Primeau, 2005).

Here, we focus on the ocean pathways that start from the seafloor and return to the surface to quantify the sequestration efficiency of the deep ocean for carbon storage. In reality, removed/captured carbon would likely be stored in the sediments, with the ocean providing an additional layer of sequestration by delaying the return of carbon back into the atmosphere. This study specifically addresses the following questions:

1. How long does it take for water at the seabed to return to the ocean surface?
2. What fraction of water remains out of contact with the atmosphere after a given time (e.g., 1,000 years)?
3. How do internal climate variability and climate change affect the return time from seabed to surface?

To answer these questions, we computed the transit-time distributions (TTDs) for water to transit from the seabed to the surface layer using monthly ocean transport matrices built from archives of the Australian Community Climate and Earth System Simulator (ACCESS-ESM1.5, Ziehn et al., 2020) submitted to the 6th phase of the Climate Model Intercomparison Project (CMIP6). TTDs were efficiently computed from an adjoint boundary propagator for the seasonally varying but annually periodic flow representing the ocean circulation climatology of the 2030s for the Shared Socioeconomic Pathway SSP3-7.0 (Arias et al., 2021; Riahi et al., 2017). Since this idealized framework cannot capture transience, we assessed the effect of climate change by the end of the century by repeating the TTD computations for the 2090s. The effect of climate variability was assessed using the 40-member ensemble of realizations of ACCESS-ESM1.5.

2. Methods

Our simulations were conducted in the idealized context of perpetually repeating a seasonally varying but annually periodic year representing the ocean circulation climatology of a given period. Although this idealized framework cannot capture the effects of drift or transience, it has multiple advantages. Chiefly, it allowed us to simulate tracers for thousands of years, well beyond the typical 1850–2100 window of CMIP6 archives. In addition, it simplified the mathematics and facilitated computing the adjoint boundary propagator required for computing the sequestration efficiency. Furthermore, it afforded us considerable computational savings, with wall times about 150× shorter than standard climate-model simulations, while retaining the features of seasonality (improving on the steady-state context of Holzer et al. (2020), Pasquier, Holzer, and Chamberlain (2024), and Pasquier, Holzer, Chamberlain, Matear, and Bindoff (2024)). Our methodological pipeline is illustrated in Figures S1 and S2 in Supporting Information S1.

2.1. Ocean Transport Matrices

We used transport matrices to leverage state-of-the-art linear solvers and efficiently compute seabed-to-surface TTDs. Briefly, a transport matrix $\mathbf{T}$ represents an ocean model's advective–eddy-diffusive transport. Multiplying $\mathbf{T}$ by a vector $\mathbf{c}$ representing tracer concentration c gives its flux-divergence, $\mathbf{u} \cdot \nabla c - \nabla \cdot \mathbf{K} \nabla c$, where $\mathbf{u}$ is the ocean model's advective velocity and $\mathbf{K}$ its diffusivity tensor.

We built transport matrices from monthly climatologies of ACCESS-ESM1.5 ocean circulation following Chamberlain et al. (2019), improving on their methodology by capturing seasonality. The horizontal advective fluxes were taken from publicly archived CMIP6 mass-transport variables, but had to be corrected for parameterized advection (taken from CSIRO archives; see Data Availability Statement for details), and the vertical fluxes were built from the seafloor up through mass conservation. The advective part of the transport-matrix was then

built using a first-order centered scheme, improving on Chamberlain et al. (2019) by reducing the severe diffusivity of their upwind scheme.

Diffusion was parameterized as a first-order scheme with constant global horizontal and vertical diffusivities ($\kappa_H = 300 \text{ m}^2 \text{ s}^{-1}$ and $\kappa_V = 3 \times 10^{-5} \text{ m}^2 \text{ s}^{-1}$), with additional vertical diffusion in the mixed layer ($\kappa_{VML} = 1 \text{ m}^2 \text{ s}^{-1}$). These constants were calibrated for our transport-matrix estimate of the water age to match the online ACCESS-ESM1.5 1850s age. However, since the ACCESS-ESM1.5 age was not properly equilibrated in the CMIP6 archives (Mannis et al., 2024), we had to run ACCESS-ESM1.5 ourselves, using Anderson Acceleration (Khaliwala, 2023, 2024) for efficient age spin-up (For details see Text S3 and Figures S3–S7 in Supporting Information S1).

To assess climate-change effects on TTDs, we built transport matrices for the 2030s and 2090s of the concentration-driven SSP3-7.0 scenario (Arias et al., 2021; Riahi et al., 2017). We chose SSP3-7.0 because its strong radiative forcing imparts substantial changes to the ocean circulation. To assess internal-climate-variability effects, we built transport matrices for the available 40 ACCESS-ESM1.5 ensemble members, which were initialized 20 years apart from a preindustrial control run (Mackallah et al., 2022; O'Neill et al., 2016; Rashid et al., 2022).

2.2. Sequestration Efficiency and Transit Time

The key metric herein is the sequestration efficiency, $\mathcal{E}(\mathbf{r}, t, \tau)$, which is the fraction of water at location $\mathbf{r}$ and time t that will not have reemerged into the surface layer after a time τ (DeVries et al., 2012). $\mathcal{E}$ is readily given by the TTD, which quantifies the probability of all the possible times for water to transit to the surface. Below, we briefly outline the mathematics and computational methodology employed, with details provided in Texts S1 and S2 in Supporting Information S1.

The TTD function is given by

$$\tau \mapsto \mathcal{G}^\dagger(\mathbf{r}, t, t + \tau), \quad (1)$$

where $\mathcal{G}^\dagger$ is a Green function—the adjoint boundary propagator—which we use to track water backwards in time from future surface injection (see Text S1 in Supporting Information S1 for details). $\mathcal{E}$ is then given by the complement of the cumulative TTD function:

$$\mathcal{E}(\mathbf{r}, t, \tau) = 1 - \int_0^\tau dt' \mathcal{G}^\dagger(\mathbf{r}, t, t + t') = \int_\tau^\infty dt' \mathcal{G}^\dagger(\mathbf{r}, t, t + t'). \quad (2)$$

For simplicity, we averaged quantities over one year. For instance, we considered the mean sequestration efficiency, $\bar{\mathcal{E}}$, given by

$$\bar{\mathcal{E}}(\mathbf{r}, \tau) \equiv \frac{1}{1 \text{ yr}} \int_{t_i}^{t_i + 1 \text{ yr}} dt \mathcal{E}(\mathbf{r}, t, \tau). \quad (3)$$

One important simplification is that $\bar{\mathcal{E}}$ only depends on transit time τ and is independent of injection time t_i , because of the yearly periodicity of our idealized framework.

Another considerable simplification owing to our idealized periodic-flow framework is for computing the TTDs. In practice, $\mathcal{G}^\dagger$ is computed as the boundary impulse response (BIR) given by $\tau \mapsto \mathcal{G}^\dagger(\mathbf{r}, t_f - \tau, t_f)$ that propagates surface injection at final time t_f backwards in time (Equations S4 and S5 in Supporting Information S1). However, this is not the same as the TTD sought here (BIR $\neq$ TTD for unsteady flow; Haine et al., 2025). Nevertheless, averaged over one period, BIR and TTD are identical (Figure S8 in Supporting Information S1). Hence, the mean adjoint boundary propagator,

$$\overline{\mathcal{G}}^{\dagger}(\mathbf{r}, \tau) \equiv \frac{1}{1 \text{ yr}} \int_{t_i}^{t_i+1 \text{ yr}} dt \mathcal{G}^{\dagger}(\mathbf{r}, t, t + \tau), \quad (4)$$

was directly used to compute the mean sequestration efficiency, as

$$\overline{\mathcal{E}}(\mathbf{r}, \tau) = \int_{\tau}^{\infty} dt \overline{\mathcal{G}}^{\dagger}(\mathbf{r}, t). \quad (5)$$

To put sequestration timescales into context, we also computed the mean transit time, $\Gamma^{\dagger}(\mathbf{r}, t)$, which is the average time for water at $(\mathbf{r}, t)$ to reemerge into the surface layer. In theory, its yearly average $\overline{\Gamma}^{\dagger}$ can be computed as the 1st moment of the mean TTD as

$$\overline{\Gamma}^{\dagger}(\mathbf{r}) \equiv \int_0^{\infty} d\tau \tau \overline{\mathcal{G}}^{\dagger}(\mathbf{r}, \tau). \quad (6)$$

However, a useful property of $\Gamma^{\dagger}$ in periodic flow is that it obeys a nonlinear equation that we directly solved with a Newton–Krylov method (Bardin et al., 2014), which is considerably more efficient than computing $\overline{\mathcal{G}}^{\dagger}$ over all transit times (details in Text S2 in Supporting Information S1).

3. Results

3.1. Sequestration Efficiency, $\overline{\mathcal{E}}$

Carbon released on the continental shelf quickly returns to the atmosphere, with more than 90% reaching the surface layer in less than 100 years ($\overline{\mathcal{E}} \leq 10\%$, Figure 1a). In contrast, on the abyssal plain away from the continental shelf, over 90% of the water has not reached the surface even after 100 years. For longer horizons, more water makes contact with the surface (Figures 1b and 1c). By 1,000 years, $\overline{\mathcal{E}}$ declines below 90% everywhere except in the North Pacific Basins.

The patterns of $\overline{\mathcal{E}}$ have the markings of the ocean circulation. The isolated abyssal waters of the North Pacific Basins provide the strongest sequestration efficiencies because these waters sit at the advective terminus of the great ocean conveyor and their return pathways are dominated by slow eddy-diffusion (Holzer & Primeau, 2006; Holzer et al., 2021). To a lesser degree, the Southwest Pacific Basin, north of the Pacific Antarctic Ridge and west of the East Pacific Rise, the Indian Ocean Basins east of the Central Indian Ridge, and the North Atlantic Basins on both sides of the Mid-Atlantic Ridge, also provide relatively strong sequestration potential. Other basins provide less efficient sequestration, with the least efficient locations near the Southern Ocean (30%–90% after 300 years, 10%–50% after 1,000 years), including the Weddell Sea, the Bellingshausen, and the Australian-Antarctic Basins south of the Southeast Indian Ridge, as well as the Crozet Basin, because connections into deep branches of the Antarctic Circumpolar Current (ACC) provide a conduit back to the surface (Toggweiler et al., 2019).

Internal climate variability effects on $\overline{\mathcal{E}}$, quantified here by the 40-member ensemble range, are generally below 20% ($\overline{\mathcal{E}}$ units). However, internal variability can be high in the Weddell Sea (>60% after 100 years and ~30% after 1,000 years) and in the North American Basin (20%–40% after 1,000 years). This is because deep waters are formed convectively in ACCESS-ESM1.5 (which is commonplace for CMIP6, Heuzé, 2021), and deep convection is a process known to exhibit decadal to multi-decadal variability (e.g., Behrens et al., 2016). This results in substantial internal variability in mixed-layer depths (Figure S9 in Supporting Information S1) and thus in seabed-to-surface pathways.

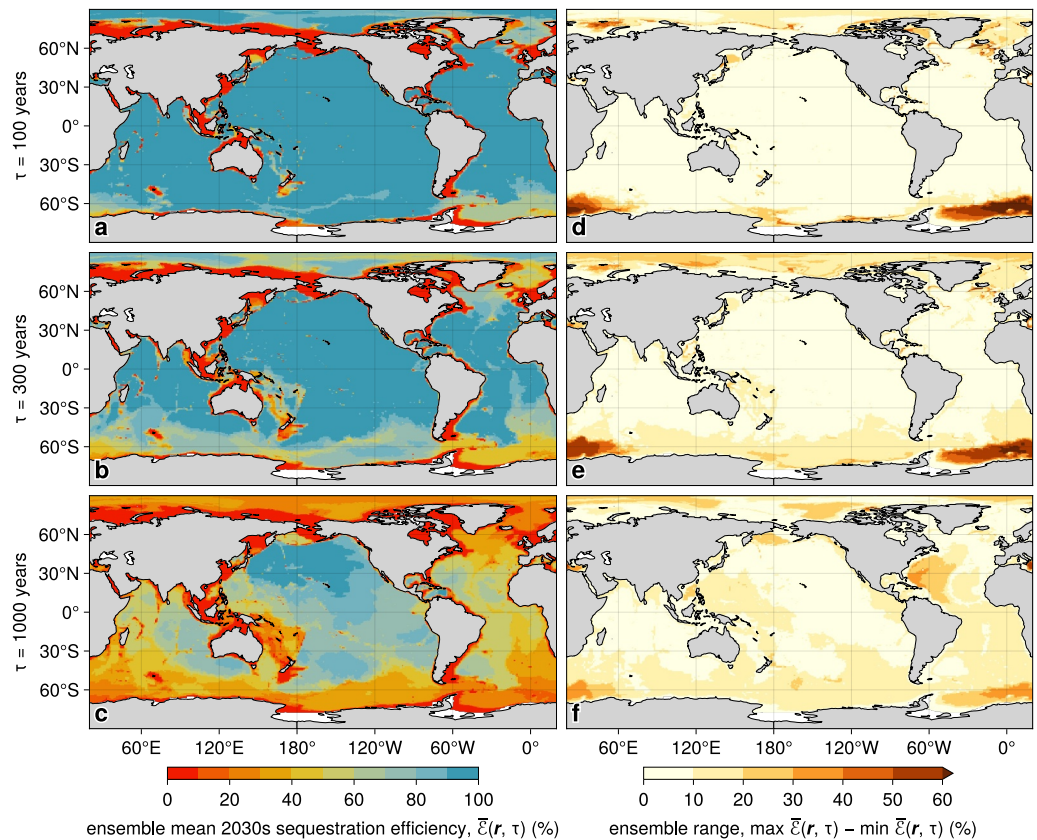


Figure 1. (a–c) Mean sequestration efficiency, $\bar{\mathcal{E}}(r, \tau)$, for seabed water after (a) $\tau = 100$ years, (b) $\tau = 300$ years, and (c) $\tau = 1000$ years, averaged over the 40-member ensemble, for the 2030s circulation. (d–f) Climate variability of (a–c) as quantified by the ensemble $\bar{\mathcal{E}}(r, \tau)$ range, that is, the ensemble maximum minus the ensemble minimum.

3.2. Characteristic Seafloor-to-Reemergence Timescales

We quantify the ocean's sequestration potential in terms of characteristic timescales of the TTD. To put these timescales in perspective, we start with the mean time to reemergence, $\bar{\Gamma}^{\dagger}$ (Figure 2a). On the abyssal plain (below 3,000 m), $\bar{\Gamma}^{\dagger}$ ranges from about 300–3,000 years. Waters of the Weddell Sea, Australian Antarctic, and Bellingshausen Basins reemerge relatively fast through ACC connections (300–1,200 years). Similarly, a fraction of waters of the Arabian Basin and the Somali Abyssal Plain upwell adiabatically into Indian Deep Water that returns to the upper ocean (Talley, 2015). Consistent with high $\bar{\mathcal{E}}$, the longest abyssal seafloor-to-reemergence times (more than 3,000 years) are found in the North Pacific Basins. We emphasize that our results are specific to ACCESS-ESM1.5, which has a relatively weak deep ocean circulation and is thus expected to overestimate transit times (see Text S3 in Supporting Information S1).

Further, we derive the quantiles of the TTD. That is, instead of charting the fraction $\bar{\mathcal{E}}$ of water that remains sequestered for a given time τ , we chart the time that a given fraction of water takes to reemerge (i.e., the time τ for a given value of $\bar{\mathcal{E}}$). For simplicity, we only consider the median ($\bar{\mathcal{E}} = 50\%$) and the 10th percentile ($\bar{\mathcal{E}} = 90\%$), that is, the time for half and 10% of the water to reemerge, respectively (Figures 2b and 2c). While their patterns are similar, the median time to reemergence is roughly $30 \pm 15\%$ shorter than the mean time to reemergence over the abyssal plain. This is because of ocean mixing, which imparts long tails on the TTDs, which skew strongly to the right (Primeau & Holzer, 2006; Waugh et al., 2006). The 10th percentile time follows similar geographical patterns and is naturally smaller with significant variations over the abyssal plain. That is, it may take a couple of centuries for 10% of the water located in the Southern Ocean abyss to reemerge, but it takes more than a millennium for water in the North Pacific Basins.

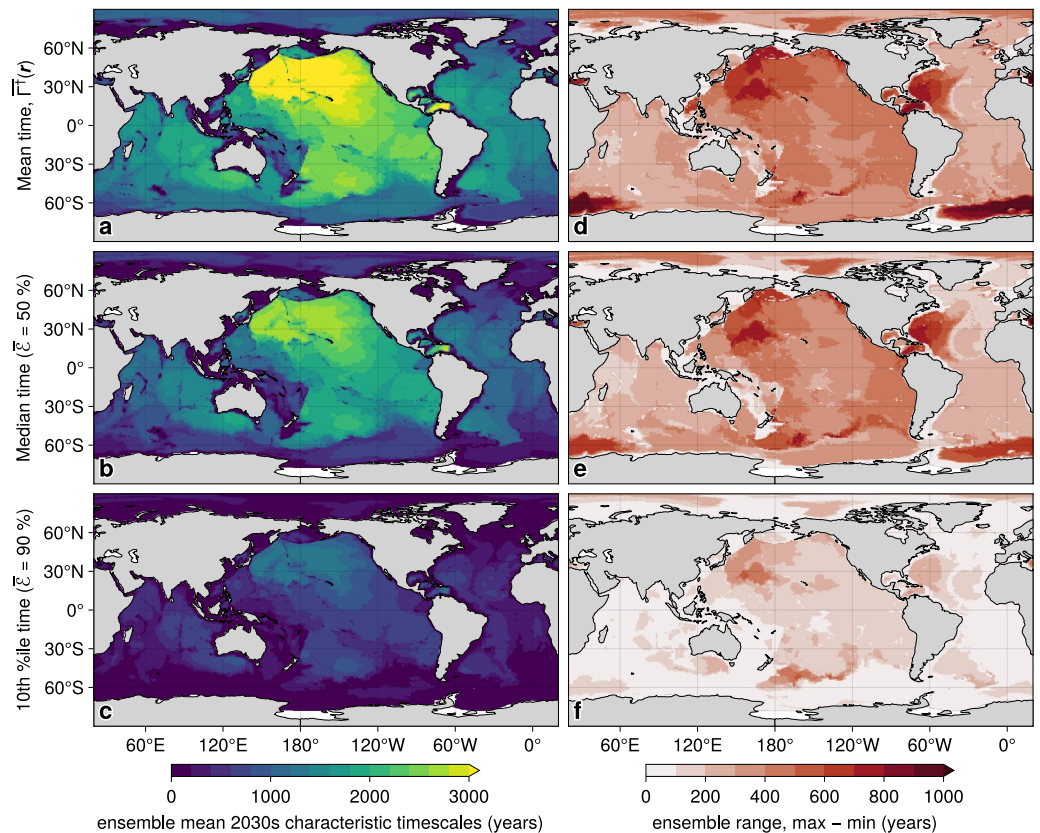


Figure 2. (a) Ensemble mean of the mean time to reemergence, $\bar{\Gamma}^{\dagger}$, at the seafloor, for the 2030s circulation. (b) As (a) but for the median time, that is, the time for half of the water to reemerge ($\bar{\mathcal{E}} = 50\%$). (c) As (b) but for the time for 10% of the water to make contact with the surface ($\bar{\mathcal{E}} = 90\%$). (d–f) Climate variability of (a–c) as quantified by the ensemble range, that is, the ensemble maximum minus the ensemble minimum.

As we did for the sequestration efficiency, we assess the effect of internal-model climate variability on reemergence timescales by looking at the ensemble range (Figures 2d–2f). We find that $\bar{\Gamma}^{\dagger}$ varies by 200–400 years or about 15%–30% of the ensemble mean over most of the global abyssal plain. The largest relative internal variability on the abyssal plain is found in the Weddell Sea Basin, where the ensemble range is larger than the ensemble mean. This is consistent with large climate variability in deep water formation and mixed layer depth in the region. Climate variability for the median time to reemergence is similar to the mean time but generally smaller. For the smaller 10th percentile time, climate variability in the Weddell Sea is almost entirely subdued (<100 years) because it takes less than 100 years for 10% of the Weddell Sea Basin water to reemerge across the 40 members.

3.3. Case Study: Injection on the Argo Abyssal Plain

As a case study, we consider the time series of sequestration for water starting 4,200 m deep on the seabed off Northwest Australia (roughly 115.5°E, 16.6°S; Figure 3a). We chose this location for its proximity to an existing offshore well, but emphasize that these time series are readily available at every ocean location within our framework, making it a useful planning tool for assessing the effectiveness of seabed carbon injection.

The TTD (Figure 3b) skews strongly to the right, with a mode at about 500–700 years and a long tail extending well beyond our 3,000 years of simulation, which is typical for TTDs in the ocean (Primeau & Holzer, 2006; Waugh et al., 2006). At about 1,600–1,900 years, the mean time to reemergence $\bar{\Gamma}^{\dagger}$ is more than twice as large as the mode.

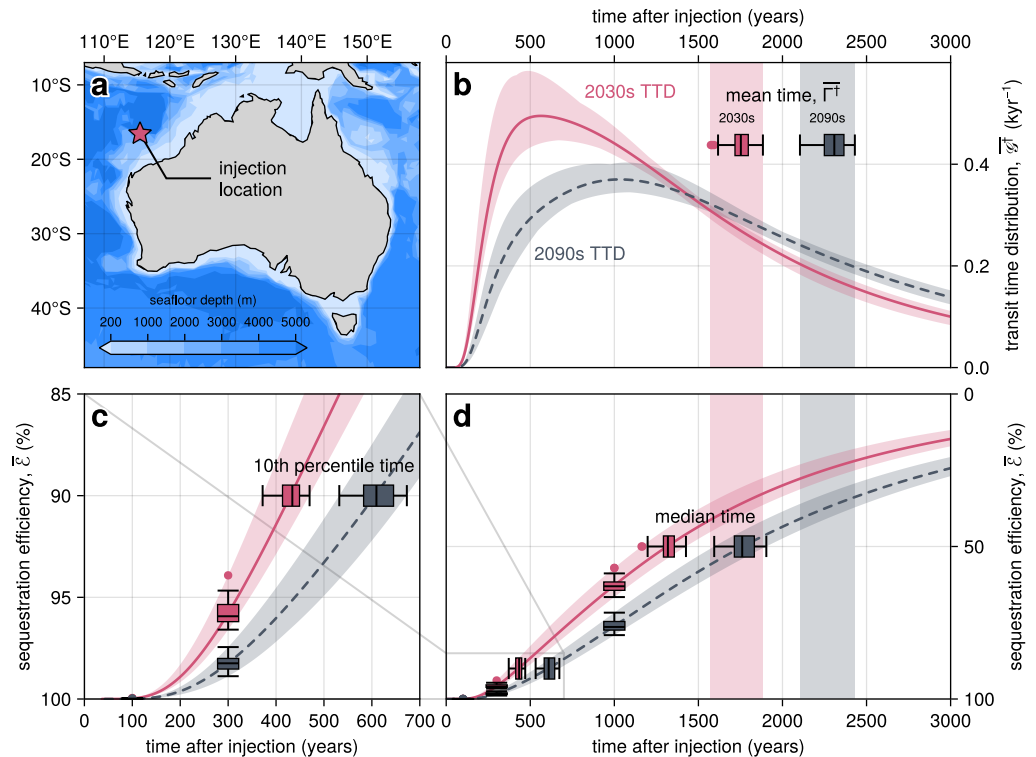


Figure 3. (a) Seafloor injection location (4,200 m depth). (b) Ensemble mean (solid lines) and ensemble range (ribbons) of the transit-time distributions for the 2030s (pink) and 2090s (gray), computed as $\bar{\Gamma}^T$. The ensemble interquartile ranges of $\bar{\Gamma}^T$ are represented as box plots and ensemble ranges as vertical bands. (c) As (b) but for $\bar{\epsilon}$. Ensemble interquartile ranges are shown for the 10th percentile time, the median time, and $\bar{\epsilon}(\tau)$ for $\tau = 100, 300, \text{ and } 1,000$ years. (d) As (c) but zoomed out to 3,000 years.

Almost no water reemerges into the surface layer in the first 150 years (Figures 3c and 3d). About 95% remains sequestered after 300 years, and 65% after 1,000 years. In terms of characteristic timescales, it takes roughly 400 years for 10% of the water to reemerge. Owing to the TTD skewness, the median time to reemergence ($\bar{\epsilon} = 50\%$), is about 15% shorter than the mean time to reemergence, $\bar{\Gamma}^T$. That is, it takes about 1,300 years for 50% of the water to reemerge. This suggests that our case-study location is a good candidate for seabed carbon injection as the ocean would provide long-term sequestration in the case of carbon leaking from the sediments.

3.4. Effect From Climate Change

Ventilation is expected to generally decrease with global warming, although the complex reorganization of the global circulation may produce counterintuitive increases (Gnanadesikan et al., 2007; Liu & Primeau, 2023). To investigate the effect of climate change on TTDs and seafloor sequestration efficiency, we repeat our analysis but for the 2090s instead of the 2030s. The expected response is a general increase in sequestration efficiency and return timescales as deep meridional overturning circulation weakens (Figures S10 and S11 in Supporting Information S1). We emphasize that our idealized simulations do not capture transient features and thus cannot accurately predict the future. Nevertheless, comparing our 2030s and 2090s simulations is useful to understand how the climate-change-induced slowdown of the circulation affects ocean sequestration.

For $\tau = 300$ and 1,000 years (Figure 4 and Figure S12 in Supporting Information S1), we find $\bar{\epsilon}$ increases by more than 30% in the Weddell Sea Basin and the Arctic, and by 10%–20% in the Atlantic. Contrastingly, we find mild decreases in the Southwest Pacific Basin north of the Pacific Antarctic Ridge, which possibly indicates emerging connections to deep branches of the ACC caused by a reorganization of the deep ocean circulation. For the short time horizon ($\tau = 100$ years; Figure S12 in Supporting Information S1), the climate-change driven increase in $\bar{\epsilon}$ is generally small (less than +10%) as it was already large in the 2030s, except in the Weddell Sea (+40%).

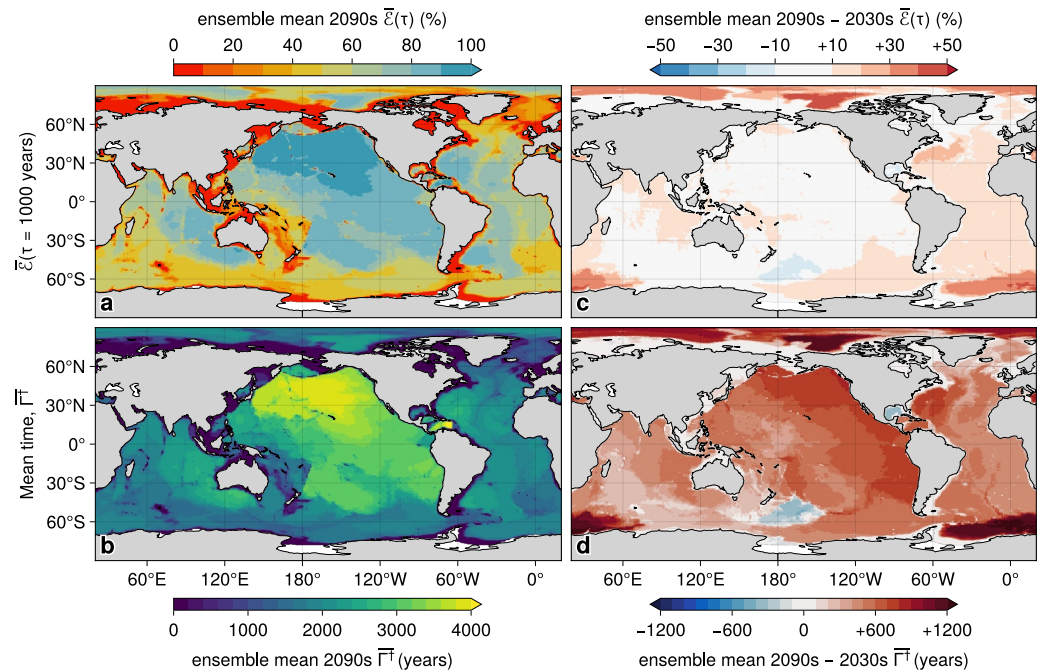


Figure 4. (a, b) As Figures 1a and 2a but for the 2090s. (c, d) Climate-change effect as quantified by the difference between the ensemble mean 2090s and 2030s.

Along with sequestration efficiency, the time to reemergence generally increases by 15%–40% for the climate-change affected 2090s ocean circulation (Figures 3 and 4, Figure S12 in Supporting Information S1). The largest increases for the mean and median time to reemergence are found in the Weddell Sea Basin and in the Arctic (+1,000–1,200 years), in the Northeast Pacific Basin (+700–800 years), and in the Northwest Atlantic Basin (+600–700 years). The increase in the median and 10th percentile times are smaller in magnitude but remain large relative to their 2030s values (+10%–50%). We note that while our analysis and the magnitude of our estimates are model dependent, the climate-change forcing, the mechanisms at play, and the relative changes in transit times should be more robust across models.

4. Conclusions and Discussion

We have introduced an idealized but computationally efficient framework to estimate the deep ocean's sequestration efficiency from existing climate-model output. Seabed-to-surface TTDs were computed in a seasonally varying annually periodic flow representing the decadal climatology of the 2030s circulation of the ACCESS-ESM1.5 model for the SSP3-7.0 scenario. We repeated this analysis across a 40-member ensemble to assess the effect of climate variability, and then for the climatology of the 2090s circulation to assess the effect of climate change.

While sequestration efficiency is generally high on the abyssal plain, we found large geographical variations across basins, imparted by recognizable patterns of the ocean circulation. The most efficient sequestration can be found in isolated waters where transport back to the surface is dominated by slow eddy-diffusion (Holzer & Primeau, 2006; Holzer et al., 2021). In contrast, abyssal waters that are swept into deep branches of the conveyor belt, which presents a relatively fast conduit back to the surface, provide the least efficient sequestration. Seabed-to-surface TTDs typically skew strongly to the right, such that it takes about 30% less time for half of the water to resurface compared to the mean reemergence time. The quantitative answers to the questions posed in this study are summarized below for the ACCESS-ESM1.5 model:

1. For abyssal basins that are swept by the conveyor belt, such as the Weddell Sea, the Bellingshausen, and the Australian-Antarctic Basins, it takes roughly 100 years for 10% and up to 600 years for 50% of the water to resurface (with a mean time of ~1000 years). Waters in isolated abyssal plains take significantly longer, for

- example, in the North Pacific Basins, where it can take up to 1,300 years for 10%, and more than 2,500 years for 50% of the water to resurface (>3000 years on average).
2. Water sitting on the abyssal plain remains well sequestered on multi-centennial timescales, while the bulk of water on the continental shelf resurfaces in less than a century. The sequestration efficiency of the abyssal plain generally remains above 90% after 100 years, above 80% after 300 years, and above 30% after 1,000 years. The North Pacific Basins provide the most efficient sequestration, with more than 90% of the water remaining below the surface after 1,000 years. The Southwest Pacific Basin, north of the Pacific Antarctic Ridge also provides efficient sequestration (>80% after 1,000 years).
 3. Climate change generally increases time to reemergence and thus sequestration efficiency, through its slowdown of the circulation. Seabed-to-surface TTDs skew further to the right, with transit taking about 30% more time (consistent with Holzer et al. (2020)), which is greater than internal climate-model variability at about 20%. This effect is particularly pronounced in the Weddell Sea where deep-water formation is dramatically reduced for our 2090s circulation, adding an average 1,200 years for seabed water to reemerge.

Our work combined and highlighted several novel conceptual and computational methods that could benefit other modeling efforts. These methods include building monthly transport matrices from climate-model archives that accurately capture the ocean transport, efficient Newton–Krylov solvers for periodic solutions, efficient forward and backward time-stepping for estimating propagators and TTDs, and Anderson Acceleration for efficient spin-up. We propose that applying some of these methods to a wide range of CMIP models could significantly help our understanding of the ocean system.

Accurate estimates of sequestration efficiency are crucial for CDR and CCS, for which the resources required are substantial (Smith et al., 2016; Sykes et al., 2020). It is therefore critical to plan these projects in regions where long-term storage is feasible. While our study clearly highlights the high sequestration potential of isolated abyssal waters of the ocean, it is specific to seabed carbon injection and ACCESS-ESM1.5. We hope that our analysis will be repeated across CMIP models and extended to other CCS/CDR techniques, such as Ocean Iron Fertilization or Ocean Alkalinity Enhancement. Our analysis could also be used beyond CSS and CDR, for example, for assessing the potential impacts of deep-sea mining on surface ecosystems through trace metals reaching the surface (e.g., Drazen et al., 2020; Weaver et al., 2022).

The importance of CMIP efforts for addressing global warming cannot be overstated. However, our work has revealed significant oversights with mass transport and water age archives, which are valuable metrics of the ocean circulation in a changing climate. We therefore urge the CMIP community to consider more systematic quality controls of these variables in future releases.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

The ocean mass transport from resolved velocity fields used (the umo, vmo variables), grid metrics (areacello, volcello), and mixed layer depth variables (mldst) are available on CMIP6 archives, for example, at <https://esgf.nci.org.au/projects/esgf-nci/>. For the unresolved Gent–McWilliams and submesoscale mass transport fields, which were not archived with CMIP6 for ACCESS-ESM1.5, we used output from CSIRO production archives of the corresponding MOM5 transport diagnostics of the ACCESS-ESM1.5 runs (using tx_trans_gm, ty_trans_gm, tx_trans_submeso, and ty_trans_submeso). The mean TTD ($\overline{g^*}$) and the mean sequestration efficiency ($\overline{\mathcal{E}}$) as plotted in the figures are available on Zenodo (Pasquier, 2025a).

The code used in this work was written in Julia (Bezanson et al., 2017), Python (Van Rossum & Drake, 1995), and MATLAB (<https://www.mathworks.com>). The Python and Julia code used to preprocess CMIP6 output data is available at <https://github.com/TMIP-code/notebooks> (Pasquier, 2025c). The Julia code used for building transport matrices from ocean model output (OceanTransportMatrixBuilder.jl, Pasquier, 2024) is available at <https://github.com/TMIP-code/OceanTransportMatrixBuilder.jl>. The Julia code for computing the adjoint boundary propagator and plotting the figures (using Makie.jl, Danisch & Krumbiegel, 2021) of this manuscript is available at <https://github.com/TMIP-code/ACCESS-TMIP> (Pasquier, 2025b). The MATLAB code for the

Anderson Acceleration algorithm used (Khatiwala, 2022, 2023, 2024) for efficient spin-up of the ACCESS-ESM1.5 water age is available at <https://github.com/briochemc/AndersonAcceleration/tree/ACCESS-ESM1-5> (Pasquier et al., 2025). The age was spun up by using the historical ACCESS-ESM1.5 model configuration (Ziehn et al., 2020) available at <https://github.com/ACCESS-NRI/access-esm1.5-configs> and using the infrastructure provided by ACCESS-NRI, which is enabled by the Australian Government's National Collaborative Research Infrastructure Strategy (NCRIS).

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